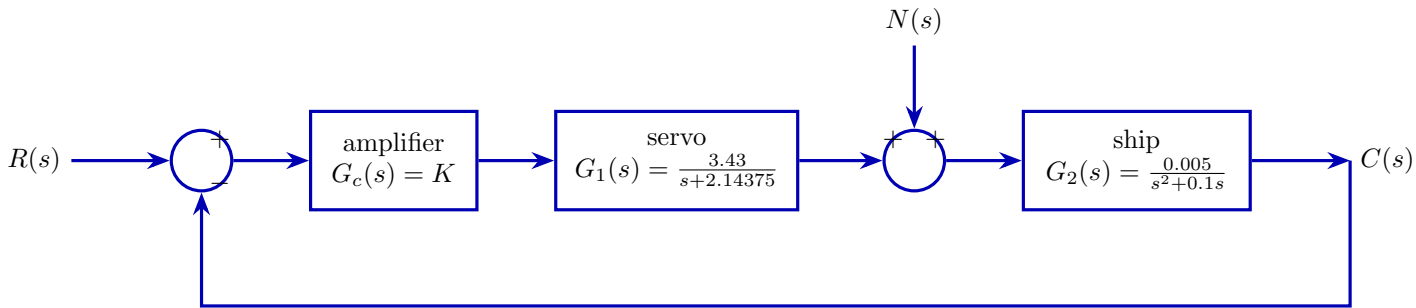




1. Use root locus to analyze how the proportional gain changes the heading-control poles.

2. Choose dominant poles at the best damping ratio and convert the locus gain back to the controller gain.

3. Compare dominant-pole design with the actual closed-loop maneuvering response.